Week 6

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. List the sample space for a chance process with equally likely outcomes. (CLO 3) **2.** Compute probabilities by counting equally likely outcomes. (CLO 3) **3.** Apply the complement rule. (CLO 3) **4.** Apply the addition rule to find P(A or B) from a two-way table. (CLO 3) **5.** Compute conditional probabilities from a two-way table. (CLO 3) **6.** Apply the multiplication rule for draws with and without replacement. (CLO 3) **7.** Distinguish disjoint events from independent events. (CLO 3)	e-Text: P.1 Probability Rules Interactive Lecture: P.1 Probability Rules Slides: P.1 Probability Rules	Participation: Regression Line – Mon: wrap-up of 2.5 Your Turn: Putting It All Together (used-car price vs. age: slope, intercept, prediction, residual) Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement Your Turn: Probabilities from a Table (job by enrollment status)	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
8. Classify a numerical summary as a parameter or a statistic. (CLO 5) **9.** Use the correct notation for parameters (μ, σ, p, ρ) and statistics ($\bar{x}$, s, $\hat{p}$, r). (CLO 5) **10.** Describe how a sampling distribution is built from repeated samples and what each dot represents. (CLO 5) **11.** Describe the center and shape of a sampling distribution. (CLO 5) **12.** Explain the standard error as the variability of a statistic. (CLO 5) **13.** Predict how the sample size affects the standard error. (CLO 5)	e-Text: 3.1 Sampling Distributions Interactive Lecture: 3.1 Sampling Distributions Slides: 3.1 Sampling Distributions	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents Your Turn: Parameter or Statistic?	Online Homework: 3.1 Sampling Distributions (due Sun Oct 11, 11:59 PM) Take-Home Quiz 4 (due Mon Oct 12, 11:59 PM) – covers 3.1–3.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus

Also this week: Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.